

Mayanka Medhe

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EDUCATION

Carnegie Mellon University

Master of Science in Robotic Systems Development

Project: VLA manipulation stack on Unitree G1, retargeting human factory video for policy pre-training (Sponsor: Nissan)

Pittsburgh, PA

Aug 2026 - May 2028

University of Tokyo

Masters in Information Science & Technology | MEXT Scholar

Tokyo, Japan

Sep 2020 - Aug 2022

Thesis: Gaze-guided dense video captioning for egocentric attention understanding (Advisor: Prof. Yoichi Sato)

Indian Institute of Technology Bombay

Bachelor of Technology with Honors in Computer Science and Engineering

Mumbai, India

Jul 2016 - Aug 2020

Thesis: Multi-Agent Path Planning and Collision Avoidance using RL (Advisor: Prof. Leena Vachhani)

Semester Exchange: Simon Fraser University, Canada

SKILLS

ML & Perception: PyTorch, VLMs (CLIP), SLAM (ORB-SLAM2), Pose Estimation, Stereo Vision, TensorRT, Transformers

Robotics: ROS/ROS2, Gazebo, MuJoCo, MoveIt, Teleoperation, Sensor Calibration, Motion Planning, Edge Deployment

Programming: Python, C++, C, Bash, SQL, Matlab, Docker, Git, CI/CD, Linux, CUDA

Spoken Languages: English, Japanese, Hindi, Marathi

PROFESSIONAL EXPERIENCE

Honda R&D

Research Engineer, Human-Robot Interaction Team

Tokyo, Japan

Sep 2022 - Aug 2026

Embodied multimodal AI and human-robot interaction for autonomous micro-mobility

- Led development of human-robot interaction system using VLMs for real-world robot control via voice intent recognition
- Deployed in public [proof-of-concept](#) with 3000+ users in unstructured environments & showcased at IFAC 2023
- Spearheaded pointing gesture recognition for vehicle stop-position estimation via stereo triangulation and 3D pose
- Developed pedestrian attribute recognition and optimized real-time object detection for edge deployment on Jetson
- Proposed text-guided map retrieval with GPS-contrastive CLIP & OCR-aware re-ranker, improving R@1 by 18.6 points
- Received Honda R&D **Executive Award** for work on multimodal HRI for next-generation mobility

Tier IV

Research Engineer, Sensing & Perception Team

Tokyo, Japan

May 2021 - Oct 2021

Object detection evaluation for autonomous driving

- Built ROS-based object detection evaluation framework flagging false positives and negatives across weather conditions

Honda Research Institute

Research Engineer, Vision Team

Wako, Japan

May 2018 - Jul 2018

Adaptive-resolution 3D volumetric mapping for bipedal robot ASIMO

- Constructed multi-resolution 3D volumetric maps from ORB-SLAM2 point clouds for robot world modeling

RESEARCH EXPERIENCE

Carnegie Mellon University

Research Assistant | Advisor: Prof. Henny Admoni

Pittsburgh, PA

Sep 2026 - Present

Interactive reward learning for out-of-distribution adaptation of visuomotor policies

- Developing multimodal human-in-the-loop reward learning for real-time policy adaptation in OOD settings

University of Tokyo

Research Assistant | Advisor: Prof. Yoichi Sato

Tokyo, Japan

Oct 2020 - May 2022

Multimodal video understanding using eye gaze and natural language

- Proposed gaze-guided dense video captioning task, integrating gaze priors into I3D features via attention maps
- Created a multimodal video dataset of gaze-aligned natural-language descriptions (2.5 hrs, 33 users, 15 languages)
- Achieved 24.4% higher action-grounding accuracy with active-viewer gaze through gaze-modulated features

PUBLICATION & PATENTS

- **Mayanka Medhe**, Naoki Hosomi. "Two-step spatially-aware and OCR-grounded map tile retrieval and point prediction for text-guided localization". Invention disclosure filed (Honda), 2026.
- **Mayanka Medhe**, Naoki Hosomi. "User localization in camera space using combined user and surrounding object information". PCT Patent Application No. PCT/JP2025/030309, 2025.
- **Mayanka Medhe**, Anirudh Reddy Kondapally, Kentaro Yamada, Hitomi Yanaka. "Improving reasoning for outdoor navigation using commonsense knowledge insertion". In SICE Annual Conference, 2023.